

Marlon Martin

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Experience

AI & Backend Engineer | *Anervea.ai* – Full Time Aug 2026 – Present

- Optimized data-quality and integration logic across 25+ collectors, resolving 48 verified defects for a pharma intelligence pipeline.
- Built the AI voice agent for a live AI-conducted interview platform, engineering streaming speech synthesis and real-time barge-in handling, physician speech mid-turn stops the AI and hands over the floor, cutting voice-response latency 68% (2.8s to 0.9s to first audio).
- Built write-time data normalizers for company names and dates, standardizing output across 25+ collectors before persistence.
- Implemented entity-resolution and deduplication logic, collapsing duplicates into single canonical records across sources.
- Built a multi-turn conversational move engine (clarify, redirect, decline, reflect) where a state machine validates every AI-proposed move and deterministically falls back on invalid output, guaranteeing safe behavior regardless of model response.
- Engineered atomic session locking (Lua-script ownership checks, 45s TTL auto-expiry) with Redis-backed state rehydration, eliminating dropped sessions on backend crash or restart across concurrent processes.

AI Engineer | *Exist Software Labs, Inc.* – Full Time Feb 2026 – July 2026

- Optimized document analysis by fine-tuning SLMs, reducing processing latency from 2 minutes to 40 seconds in the PoC.
- Built a Hybrid RAG pipeline using Azure AI Search and reranking, cutting analysis time by 65% for faster retrieval.
- Implemented signature verification using feature extraction and verification, achieving 80% accuracy without Neural Networks.
- Deployed an AI Governance framework with custom guardrails, ensuring 100% deterministic security across LLM providers.
- Developed a CV pipeline using PyTorch, YOLO, and SAM3 to enable vector-based signature verification for claims.
- Architected a self-hosted OCR service using PaddleOCR with swappable backends, reducing costs and eliminating vendor lock-in.
- Developed an internal procurement scraper for government bidding platforms, ranking IT bids via FAISS matching.
- Evaluated enterprise AI gateways, building an internal knowledge base for company strict infrastructure compliance.

Backend Engineer | *Freelance* – Remote, Contractual June 2024 – Jan 2026

- Architected lending infrastructure using Domain Driven Design and state machines for 100% financial lifecycle traceability.
- Reduced deployment latency by 87%, cutting global build-to-deploy pipelines from 41 minutes to under 5 minutes.
- Engineered hybrid RBAC/PBAC with atomic transactions, ensuring 100% data integrity and granular access control.
- Developed automated TDD suites, maintaining high-leverage reliability and 95%+ code coverage across services.
- Optimized CI/CD via Turborepo and remote caching, eliminating 90% of redundant build tasks for faster delivery.

Lead Software Engineer | *Binspire* – Full-time Oct 2023 – Dec 2025

- Directed the Software and AI engineering lifecycle, from cost-finance modeling to multi-node deployment for Smart-City solutions.
- Reduced module startup time by 90%, bringing system initialization from over a minute to under 10 seconds.
- Implemented a vision pipeline with QR verification, ensuring 100% log accountability and data integrity.
- Architected an autonomous Agentic Engine using custom MCP servers to expose IoT data for multi-model orchestration.
- Engineered an IoT fleet-management system using Raspberry Pi and MQTT for high-performance Arcovia City operations.
- Optimized collection routes via agentic reasoning, improving logistics efficiency for urban waste management.

Projects

Regtrace | *TypeScript, CI, DevOps, LLM Evaluation Framework* June 2026

- Built a CLI framework for evaluating and benchmarking LLM outputs across factuality, format, tone, and regression; integrated quality gates and golden-set baselines into CI/CD pipelines to automatically detect output regressions before deployment.

Ollamaduct | *TypeScript, AI Gateway, CLI, Ollama* March 2026

- Local-first AI gateway for Ollama with multi-tenant workspaces, semantic caching (95% match rate), and PII redaction for 25+ entity types, reaching 380+ weekly npm downloads.

Workflow | *TypeScript, AI Agent, LangGraph, MCP, Multi-Agent Systems* November 2025

- Agentic Waste Engine using a Multi-Agent System to convert rigid collection schedules into a demand-responsive network, with a custom MCP server bridging the Binspire API to LLMs.

Activities

14th IT Skills Olympics | *Participant (Team Leader)* November 2025

- Led IoT-enabled agentic waste management engine project to 1st Runner-Up at the 14th IT Skills Olympics.

Competitive Programming | *Participant* February 2023

- Ranked among top performers in university-level competitive programming, solving 3/4 problem sets in Java as a first-year student.
- Regularly compete on LeetCode, HackerRank, Codeforces, and other online platforms, focusing on algorithms, data structures, and time-constrained problem-solving.

Open-Source | *Developer & Contributor* January 2023 – Present

- Enhanced functionality, documentation, and accessibility for global and local projects, including LangChain.js, OSSPH, and BetterGov.ph, by collaborating with community developers.

Skills

Programming Languages: C, C++, Java, SQL, Python, JavaScript, TypeScript

Libraries & Frameworks: LangGraph, LangChain, Haystack, MCP, N8N, PyTorch, OpenCV, Roboflow, UiPath, Hono.js, FastApi, FAISS

Data & Infrastructure: AWS, Azure, GCP, Cloudflare, Docker, Nginx, HiveMQ, Langfuse, LangSmith, Hugging Face, MongoDB

Education

Pamantasan ng Lungsod ng Pasig

June 2022 – June 2026

Bachelor of Science in Information Technology

Awards:

- President's List Academic Achievement Awardee

Rizal High School

March 2020 – March 2022

ICT Strand – Computer Programming

Awards:

- 1st Overall – Best System Presenter

- Dean's List Academic Achievement Awardee
- 1st Runner Up 14th IT Skills Olympics (IoT Category)
- Best Thesis Innovation in Internet of Things – Academic Special Awardee
- Outstanding Student Athlete – Academic Special Awardee

- 1st Overall – Best System Design
- 1st Overall – Best System Functionality
- 1st Overall – Best System
- Top Programmer of the Class – Academic Special Awardee